BridgeFinder: Finding Research Collaborators Across Communities at UIUC

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Abstract

Students often look for research opportunities through people they already know. BridgeFinder helps them look beyond those circles. A student selects a familiar researcher, explores a map of collaboration groups, and sees suggested researchers with related work and possible introduction paths. At midterm, we have built the web prototype, analyzed two UIUC publication datasets, and gathered feedback from five students. A test using past collaborations finds that our recommendations predict later cross-community collaborations better than shared-contact ranking, although the gains fall short of our goals. The next step is to test whether students find these recommendations relevant, unfamiliar, and useful for contacting someone.

1 Objective

A student who enjoyed a computing class might want to apply those skills in biology or agriculture. The difficult part is often knowing whom to approach. The student may know their instructor but have few connections outside the department. Searching by keyword can return relevant names without explaining how to start a conversation.

BridgeFinder finds researchers with related interests outside a familiar researcher's collaboration group and shows who might connect them. We focus on UIUC undergraduate and early-stage graduate students looking for research opportunities beyond their current department. A student can begin with an instructor or researcher whose work they know, without needing publications of their own.

The main design challenge is to balance relevance with discovery. A suggestion should match the student's interests while adding someone they would be unlikely to find through their usual contacts. It should also explain a practical next step. At midterm, we have a working three-screen app, a pipeline for building the collaboration graph, tests on two publication datasets, and one round of student feedback. Our results show promise for finding later collaborators; whether the app helps students find mentors remains an open question.

2 Related Work

Search and visual discovery. Illinois Experts [11] and Google Scholar help users find expertise through profiles and publications. They are useful when a student knows what to search for, but a list of relevant names does not by itself explain whom the student could ask for an introduction. Connected Papers and ResearchRabbit make related literature easier to explore through visual maps. We borrow that visual approach, but organize the map around researchers and their collaborations rather than papers. Our main addition is to connect a recommendation to a possible introduction path.

Predicting collaborators. Liben-Nowell and Kleinberg [7] study how network structure predicts future links. One useful measure is Adamic–Adar, which favors people with shared contacts. This gives an understandable way to suggest likely collaborators, but can keep recommendations close to familiar circles. Tang et al. [10] study cross-domain collaboration recommendations by combining evidence about research interests and relationships. That work supports using more than topic matching alone. Our project asks an additional question: can students understand and act on the connection we show? We therefore evaluate both ranking quality and students' judgments of the recommendations.

Bridging research groups. Granovetter's work on weak ties [6] explains how contacts outside a close circle can bring new information. Burt's work on structural holes [2, 3] describes the opportunities available to people who connect otherwise separate groups. These ideas motivate BridgeFinder, but do not guarantee that any particular missing collaboration would be valuable. Dahlander and McFarland's study of research organizations [4] also shows why intellectual similarity and organizational context matter for collaboration. Our graph captures shared publications and topics; it cannot capture every personal or institutional reason a collaboration succeeds.

3 Course Concepts and Network Analysis

3.1 Building the collaboration graph

We use OpenAlex [9] publications from 2015–2024. Each node represents an author affiliated with UIUC on a collected paper. Two authors share an edge if they wrote a paper together. We keep papers with at least two UIUC authors and omit collaborators outside UIUC.

We collected a pilot sample of 5,000 works and a larger sample of 25,000 works. After keeping papers with multiple UIUC authors, these contain 2,493 and 13,229 papers, respectively. They are samples of campus research, not a complete directory. Table 1 describes the graphs built from the earlier papers, which we use to make predictions.

Table 1: Collaboration graphs from 2015–2019 papers. A community is a group found from collaboration patterns, not a department.

	Pilot	Larger sample
Authors with collaboration edges	3,642	9,042
Co-authorship edges	16,408	36,748
Detected communities	344	454
Modularity	0.949	0.925

We give each shared paper a weight of $1/(a-1)$, where a is the number of UIUC authors on that paper, and add those weights across papers. This reduces the influence of large teams: one paper with many authors should not count as strongly as repeated collaboration in a small group.

3.2 Finding communities and possible bridges

We use Louvain community detection [1] to find groups with many collaborations inside the group and fewer outside it. Louvain seeks high *modularity*, which measures how strongly edges fall within groups compared with a network with similar node strengths. We chose it over Girvan–Newman [5], which repeatedly removes edges that lie on many shortest paths, because that repeated computation is costly on graphs of this size.

The high modularity in Table 1 suggests that the observed collaborations are concentrated in separate groups. This supports looking across group boundaries. However, some groups are in completely disconnected parts of the graph, which raises modularity even without a meaningful disciplinary divide. We label communities by number rather than treating them as verified departments.

This is where structural holes and brokerage enter the app. We consider researchers in a different community who have not already co-authored with the selected researcher. A missing edge is a possible collaboration to explore, not proof that the two people do not know each other. A shared co-author provides a possible introduction; a longer path may require more intermediaries. These are observed publication paths, not measurements of friendship or trust.

3.3 Ranking the suggestions

For selected researcher u and candidate v, our score is

$$\text{Score}(u, v) = 0.60S + 0.12H + 0.12B + 0.16P. \quad (1)$$

The four terms describe different reasons to recommend someone:

- **Topic match (S):** the fraction of shared topics among the two researchers' combined topic sets (Jaccard similarity). Each set contains up to ten topics from the researcher's earlier papers.
- **Autonomy (H):** one minus Burt's network constraint [2], bounded below by zero. Constraint measures how much a person relies on contacts who also connect to one another. Autonomy is higher when those contacts are less concentrated in the same circle. Nodes with at most one neighbor receive zero.
- **Connection to the starting group (B):** the fraction of the candidate's collaborators who belong to u's community.
- **Short introduction route (P):** 1 for a two-edge path, 0.5 for a three-edge path without a shorter one, and 0.1 when neither is found.

These are fixed design choices, not learned weights or probabilities of success. Topic match receives the most weight so that network position does not overwhelm relevance. We also test a simpler Topic+Path version to see whether the extra network features help. Without the network analysis, the app would lose its community filter and introduction paths and become a topic search tool.

4 Design and Implementation

4.1 From a familiar person to a new contact

The app follows the proposal's three-screen flow, with a starting-person selector added to help students who have no publications:

1. Pick a familiar researcher. The student searches a list of 50 starting researchers, chosen automatically from highly connected authors across the largest communities. The list shows names and research topics. These are historical author records; students need to check current appointments and availability.

2. Explore the map. The map shows the selected researcher, recommended candidates, and people along their collaboration paths. Solid lines show observed co-authorships. Orange dashed lines show recommended

cross-community connections that are not existing co-authorships. This gives the student context for a suggestion, rather than only a ranked name.

3. Open a researcher card. Clicking a dashed line opens the candidate's card, with topics, example papers, network scores, and the observed introduction path. The path appears first because students in our initial feedback round looked for whom to contact before reading the details. Figure 1 shows the map and one resulting card.

4.2 What works now

Python prepares the graph and recommendations; a static web page renders them with D3. This keeps graph calculations out of the student's browsing session. The app, prediction test, and study cards use the same graph and ranking code, so the model we evaluate is the one displayed in the interface.

Cards show only paths found in the data, with at most two intermediaries. If no path is found within that limit, the app says so rather than inventing a contact. An outreach button opens an editable message template but does not send email. Automated checks cover the displayed rankings and route edges, and a browser walkthrough confirms that the main flow works. These checks establish that the prototype runs; the next user round will test whether people understand it.

5 Users and Feedback

Our target users are UIUC students who want to explore research beyond their current department. They are a useful initial population because they may have subject knowledge from classes but few research contacts. We gathered early feedback from five students in individual, 30-minute sessions, during which they talked through their use of the prototype: three CS/CE undergraduate juniors or seniors and two graduate students from Bioengineering and Information Sciences. This convenience sample covers both computing and application areas, but is too small to represent the campus population. Its purpose was to find interface problems [8], not to establish statistical differences between recommendation methods.

Three observations led to design changes:

Students needed a way to start. Students without publications could not use themselves as the starting node. We added a searchable list of researchers so they could begin with an instructor or someone whose work they knew.

Dashed lines were ambiguous. Students read them as weak existing collaborations. We added a legend that separates observed co-authorships from suggested connections. The next round will check whether students can explain that difference without help.

The introduction mattered before the score. Students looked for who might introduce them before reading papers and topics. They also read a low constraint score as a poor recommendation. We moved the introduction path to the top of the card and displayed autonomy as a positive gauge. We still need to check whether this makes the explanation easier to use.

These are qualitative findings from the authors' sessions; we have not measured how common each issue was or whether the changes resolved it. The next study will include students from more disciplines and compare their judgments across recommendation methods.

6 Evaluation

6.1 Predicting later collaborations

We built the graph and topic profiles using only 2015–2019 papers, then asked whether the recommended researchers actually became new cross-community co-authors in 2020–2024. We evaluated 180 starting researchers in the pilot sample and 1,385 in the larger sample. Each had at least one eligible new collaboration in the later period. This excludes researchers who formed no such ties, so the results do not describe every researcher or student.

All methods rank the same candidates. Topic-Only tests whether topic matching is enough; Adamic–Adar tests shared contacts. Preferential attachment favors well-connected researchers, and betweenness favors researchers who lie on many shortest paths. Topic+Path tests a simpler version of our model.

We report **precision at five (P@5)**, the fraction of the first five suggestions that became collaborators, because students should not have to inspect a long list. We also report **NDCG@10**, which gives more credit when actual collaborators appear near the top of the first ten results. Both are averaged across starting researchers. Table 2 reports the current implementation's results.

Table 2: Prediction of new 2020–2024 cross-community collaborations. Higher values are better.

	Pilot		Larger sample	
Method	P@5	NDCG@10	P@5	NDCG@10
Preferential attach.	.0011	.0011	.0088	.0133
Betweenness	.0011	.0211	.0078	.0148
Topic-Only	.0044	.0274	.0208	.0402
Adamic–Adar	.0056	.0297	.0295	.0515
Topic+Path	.0133	.0615	.0344	.0620
BridgeFinder	.0133	.0514	.0351	.0674

On the larger sample, BridgeFinder improves P@5 from 0.0295 to 0.0351 over Adamic–Adar, a relative gain of about 19%. NDCG@10 rises from 0.0515 to 0.0674. However, this falls short of our targets of doubling P@5 and adding 0.05 to NDCG@10. Precision is also low in absolute terms: only about 3.5% of the top-five suggestions become observed collaborators.

The simpler Topic+Path model has the same pilot P@5 as BridgeFinder and better pilot NDCG@10, while the full model performs better on the larger sample. The added

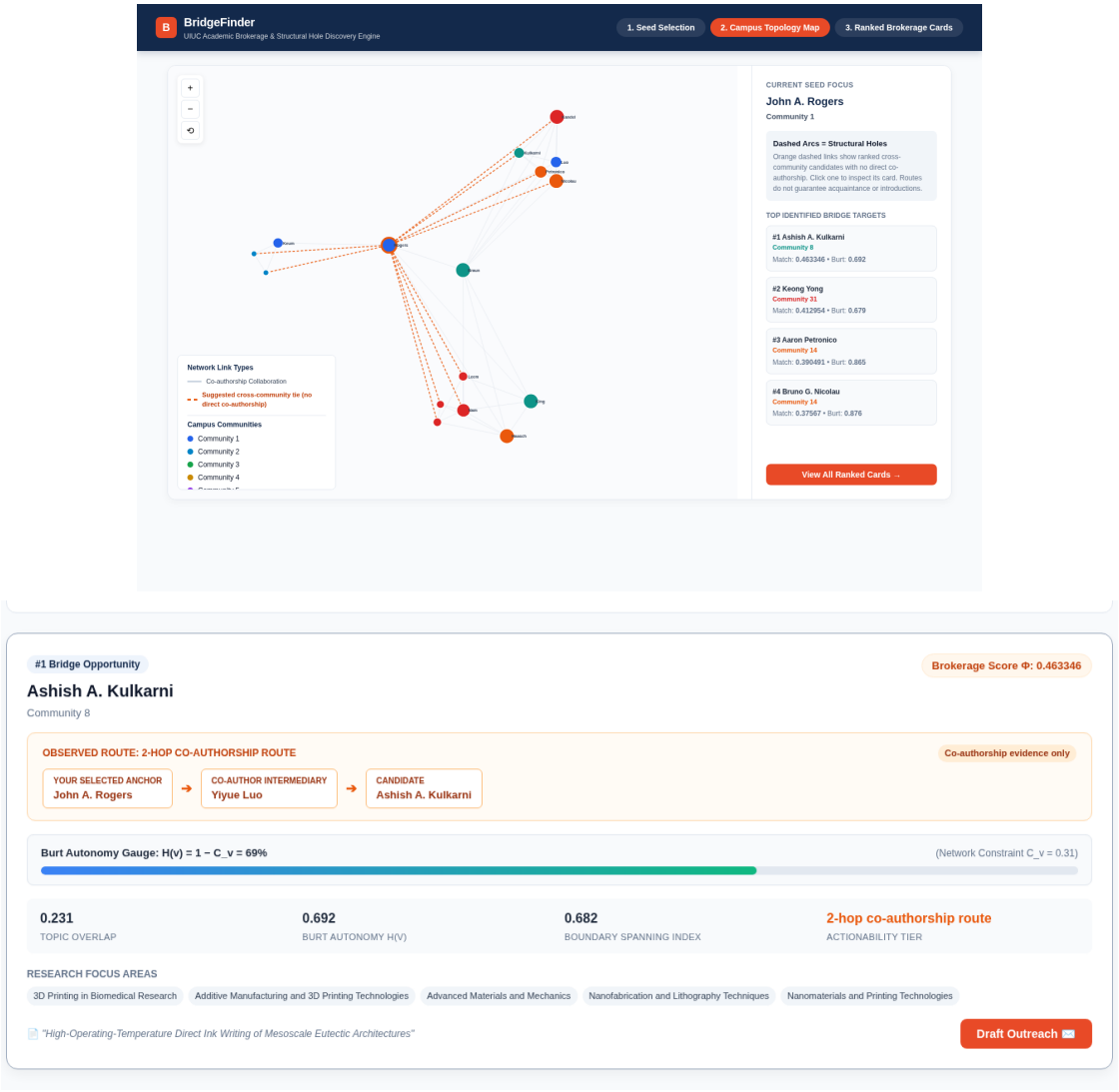


Figure 1: The implemented prototype. Top: selecting John A. Rogers shows suggestions outside his collaboration community. Dashed orange links are suggested collaborations, while solid lines show observed collaborations. Bottom: opening Ashish A. Kulkarni's card shows a possible route through Yiyue Luo, alongside shared topics and example work. A co-authorship path suggests whom to ask; it does not guarantee an introduction.

features therefore do not help consistently. We have not yet estimated uncertainty around these differences. The results suggest that network information is useful, but do not establish reliable gains across settings or show that the app helps students find mentors.

6.2 Citation impact

We also compared later papers involving authors from different training communities with papers whose known authors were in one community. This analysis separately excludes papers with more than 25 retained UIUC authors to reduce the influence of large teams.

Of 6,231 later papers, 1,071 were cross-community, 4,183 were within-community, and 977 could not be classified because none of their authors appeared in the training graph. Mean citations were 64.86 and 61.60, respectively, a ratio of 1.053. The medians were 36 and 37. A two-sided Mann–Whitney test gave $p = 0.0601$.

Our target was a mean ratio above 1.10 and a statistically significant distributional difference ($p < 0.05$); neither was met. This test also does not account for field, publication age, or repeated teams. We therefore cannot conclude that crossing communities leads to more influential work.

6.3 Student evaluation

The next round will recruit 15 students, aiming for three each from CS, ECE, Bioengineering, Crop Sciences, and Information Sciences. This broadens the first round beyond computing. It remains an exploratory study, not a representative campus survey.

Each participant will first use the app to find a candidate

and explain the suggested route. They will then choose a starting researcher and rate up to 15 shuffled cards: the top five from BridgeFinder, Adamic–Adar, and Topic-Only. Method labels will be hidden. A candidate selected by more than one method will appear once, with the rating counted for each method that selected it.

We will measure three kinds of success:

- **Relevance and discovery.** Participants rate relevance and novelty from 1 to 7. We target mean scores of at least 5 for each. Our main comparison is the share of cards rated at least 5 on both scales: BridgeFinder should exceed Adamic–Adar by 25 percentage points. Topic-Only provides a second comparison.
- **Willingness to act.** Participants rate how realistic it feels to contact the candidate, also from 1 to 7. We target at least 5 and an improvement over Topic-Only, testing whether the path adds practical value.
- **Usability and understanding.** At least 12 of 15 participants should complete the task without help and correctly distinguish observed ties from suggested connections. We will also use the standard ten-question System Usability Scale (SUS), targeting a mean of at least 80 out of 100.

We will compare methods using each participant's average ratings, rather than treating all cards as independent observations. The main joint relevance-and-novelty comparison will use a paired Wilcoxon test at $p < 0.05$, with differences and uncertainty intervals reported. Other comparisons will be exploratory and use Holm correction for multiple tests. The rating interface, SUS questionnaire, export, and analysis scripts are implemented and tested. The comparative study itself has not yet been run.

7 Limitations and Next Steps

The prototype can show a student where a relevant researcher sits in a collaboration network and suggest whom to ask about an introduction. It cannot tell whether that researcher currently takes students or whether a co-author would be willing to introduce them. Its historical UIUC sample also misses informal relationships, collaborators outside UIUC, and some publications.

The main question for the next round is whether the map and explanations help students make a useful choice. We will run the 15-student study in November, revise the interface based on observed misunderstandings, and test any scoring changes on separate validation data before the final evaluation. The final report will combine those user findings with the ranking results. So far, we have demonstrated a working system and modest prediction gains; useful mentoring connections remain the outcome we want to establish.

Contributions and AI Use

Evan Lin worked on data collection, the prediction test, and the campus map. Eric Liu worked on scoring, citation analysis, study tools, and the formative sessions. Both contributed to the report. Claude assisted with early development and drafting; OpenAI Codex assisted with implementation checks, tests, and report revision. No participant responses or quotations were generated.

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